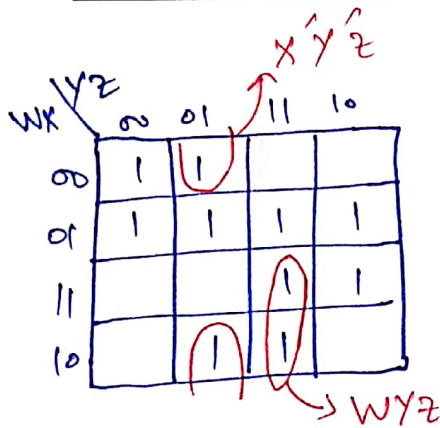
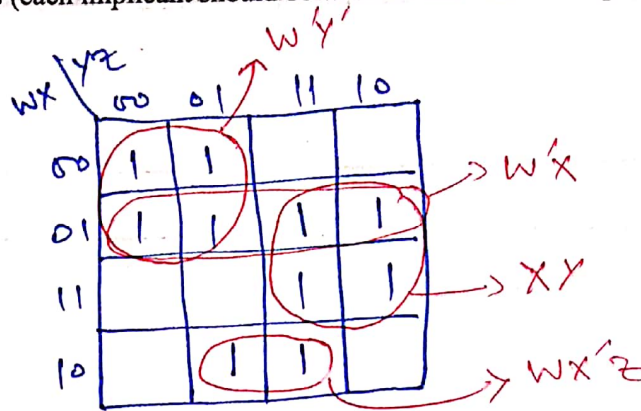


Question: Consider the following Boolean function.

$$F(W, X, Y, Z) = \sum(0, 1, 4, 5, 6, 7, 9, 11, 14, 15)$$

- a. List all the prime implicants (each implicant should be written in the form of a product term). [4]

yz \ wx	00	01	11	10
00	m0	m1	m3	m2
01	m4	m5	m7	m6
11	m12	m13	m15	m14
10	m8	m9	m11	m10



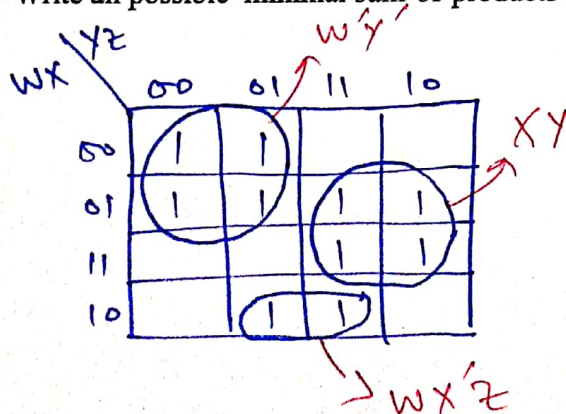
Prime Implicants are:-

$$w'y', w'x, xy, wx'z, x'y'z, wyz$$

- b. List all of the essential prime implicants (again each implicant should be in the form of a product term) [3]

$$w'y', xy,$$

- c. Write all possible minimal sum-of-products equations for this function. [3]



$$F = w'y' + xy + wx'z$$

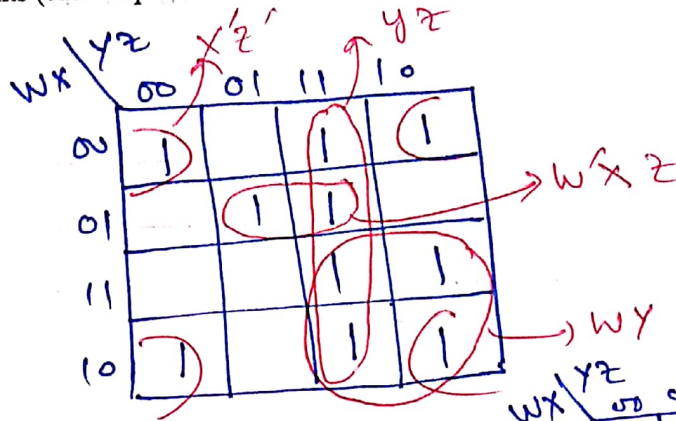
Name: _____ Roll No: _____ Section: _____

Question: Consider the following Boolean function.

$$F(W, X, Y, Z) = \sum(0, 2, 3, 5, 7, 8, 10, 11, 14, 15)$$

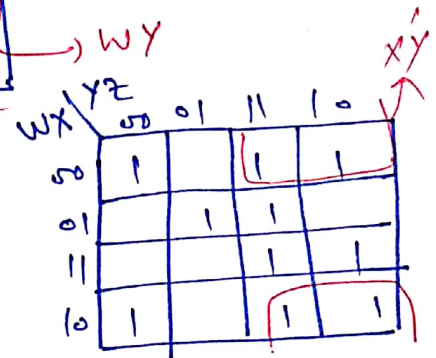
a. List all the prime implicants (each implicant should be written in the form of a product term). [4]

yz \ wx	00	01	11	10
00	m0	m1	m3	m2
01	m4	m5	m7	m6
11	m12	m13	m15	m14
10	m8	m9	m11	m10



Prime Implicants are:-

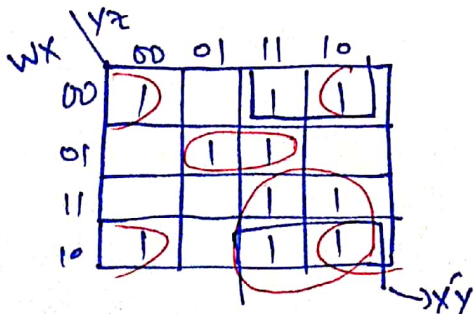
$$X'Z', YZ, WXZ, WY, XY$$



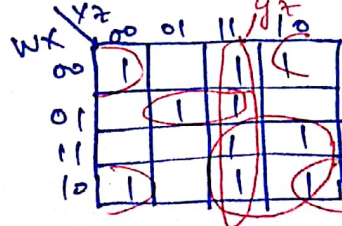
b. List all of the essential prime implicants (again each implicant should be in the form of a product term) [3]

$$X'Z', WXZ, WY$$

c. Write all possible minimal sum-of-products equations for this function. [3]



$$F = X'Z' + WXZ + WY + X'Y \rightarrow \text{1st Solution.}$$



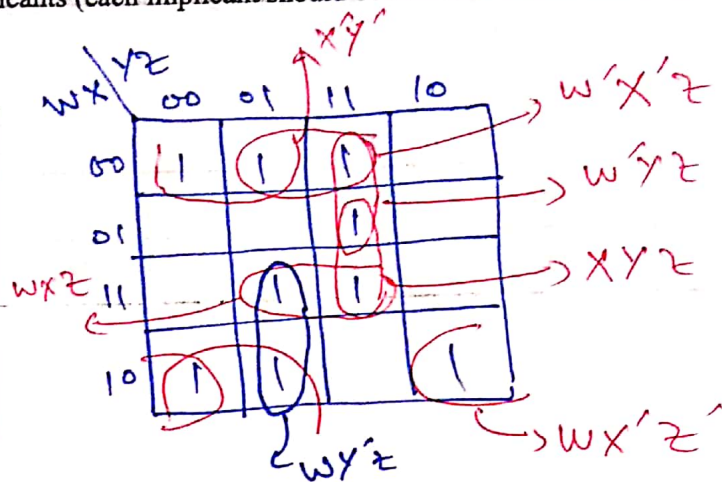
$$F = X'Z' + WXZ + WY + YZ \rightarrow \text{2nd Solution.}$$

Question: Consider the following Boolean function.

$$F(W, X, Y, Z) = \sum(0, 1, 3, 7, 8, 9, 10, 13, 15)$$

- a. List all the prime implicants (each implicant should be written in the form of a product term). [4]

yz \ wx	00	01	11	10
00	m0	m1	m3	m2
01	m4	m5	m7	m6
11	m12	m13	m15	m14
10	m8	m9	m11	m10



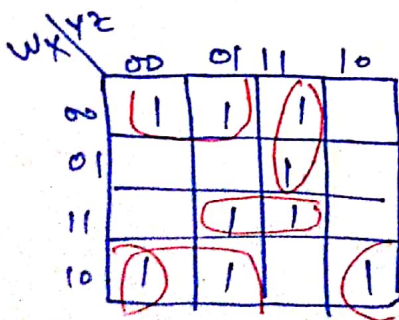
Prime Implicants are:-

$$x'y'z, w'x'z, w'yz, xyz, wxz, wx'z', w'yz$$

- b. List all of the essential prime implicants (again each implicant should be in the form of a product term) [3]

$$x'y'z, wx'z'$$

- c. Write all possible minimal sum-of-products equations for this function. [3]



$$F = x'y'z + wx'z' + w'yz + wxz$$

Name: _____

Roll No: _____

Section: _____

Question: Consider the following Boolean function.

$$F(W, X, Y, Z) = \sum(1, 3, 6, 7, 8, 9, 12, 13, 14, 15)$$

- a. List all the prime implicants (each implicant should be written in the form of a product term). [4]

yz \ wx	00	01	11	10
00	m0	m1	m3	m2
01	m4	m5	m7	m6
11	m12	m13	m15	m14
10	m8	m9	m11	m10

Prime Implicants are:-

$$w'y'z', w'yz', xy, wx, wy'$$

- b. List all of the essential prime implicants (again each implicant should be in the form of a product term) [3]

$$w'y'z', w'yz', xy, wy'$$

- c. Write all possible minimal sum-of-products equations for this function. [3]

$$F = w'y'z' + w'yz' + xy + wy'$$

Name: _____

Solution

Roll No: _____

Section: _____

Question: Consider the following Boolean function.

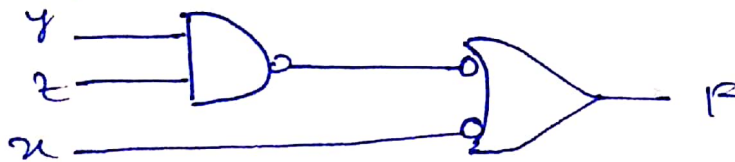
$$x'y' + yz + x'yz'$$

- a. Simplify the Boolean function using three variable k-map: [5]

x \ yz				
	00	01	11	10
0	1	1	1	1
1			1	

$$F = x' + yz$$

- b. Implement the simplified function obtained in part 'a' using two level NAND gate circuit. [5]



Question: Consider the following Boolean function.

$$xy + x'y'z' + x'yz'$$

- a. Simplify the Boolean function using three variable k-map: [5]

x \ yz	00	01	11	10
	0	1	1	0
0	1			1
1			1	1

$$F = x'z' + xy$$

- b. Implement the simplified function obtained in part 'a' using two level NAND gate circuit. [5]

